Math 1200 Lecture 1

Sep 3 2026

Topics: 12.1-12.3: 3d space, vectors, dot product Names:_____

Group work 1 Find the distance between the point $P(1, 3, 3)$ and the center of the sphere given by the equation $(x-2)^2 + (y-1)^2 + (z+1)^2 = 9$. Try to draw both the point and the sphere in 3d space.

Group work 2 Graph $z = x^2$ in $\mathbb{R}^3$. What about the region $z \geq x^2$?

Group work 3 Suppose we are given points $P(2, -4, 4)$ and $Q(4, -3, 6)$ in $\mathbb{R}^3$.

(a) Find a unit vector $\vec{u}$ pointing in the same direction at $\vec{PQ}$.

(b) Express $\vec{u}$ using $\mathbf{i}, \mathbf{j}, \mathbf{k}$.

Group work 4 Consider the vectors $\vec{v} = \langle 1, 2, 3 \rangle$ and $\vec{w} = \langle 2, -1, -1 \rangle$. Is the angle between them acute, obtuse, or are they orthogonal?